PUBLIC

Integrating SAP BTP DMS Content Server with S/4HANA

Document Owner:

SAP Enterprise Cloud Services - Service Engineering

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Disclaimer: The processes and details described in this document apply exclusively to SAP Enterprise Cloud Services (ECS). ECS does not provide a managed service for this integration. The information is intended solely to guide customers in configuring the BTP Document Management Service (DMS) Content Server integration with S/4HANA within an ECS-hosted landscape. The processes and details as described in this document are only valid for SAP Enterprise Cloud Services operated by SAP as the delivery organization. These processes and details may be different if the services are delivered by an SAP Partner / supplier.

SAP Enterprise Cloud Services

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1. Introduction

1. Overview

SAP Business Technology Platform (BTP) Document Management Service (DMS) is a cloud-native service that enables organizations to store, manage, and retrieve business documents in a secure, centralized repository. It is designed to support modern enterprise applications, providing scalability, version control, and robust access management, without requiring on-premise content servers.

2. Key Features

- Centralized Cloud Repository: Store all enterprise documents in one secure location.
- Version Management: Track and manage multiple versions of each document.
- Role-Based Access Control: Ensure only authorized users can access or modify documents.
- Metadata & Classification: Tag and categorize documents for easy search and retrieval.
- Integration Options: Supports REST APIs, application connectors, and Fiori apps for programmatic or user-driven document operations.
- Audit & Compliance: Maintain logs of uploads, edits, and access events for regulatory compliance.

3. Document Management Service – Options

Application Option: This is a SaaS offering, it provides a standalone, ready-to-use web application with a user interface for managing documents. It includes functionalities like creating, editing, downloading, and deleting files and folders, navigating hierarchies, viewing various file types, and managing document versions. It is a complete solution for document management without requiring custom development.

Integration Option: This is a PaaS offering. It provides APIs and reusable UI components that allow users to integrate document management capabilities into their existing business applications. It focuses on enabling developers to build custom solutions and embed DMS functionalities within other applications, offering flexibility and control over the user experience.

Repository Option: Repository Option isn't an independent offering. You can use it for storage purposes along with Integration Option or Application Option. You can also specify the storage location where your documents are saved with the Document Management Service. You can either use our own repository option and store the data in the SAP cloud data centers, or you can use your own CMIS-compliant **repositor**ies, such as OpenText or SAP S/4HANA DMS.

2. Integration with S/4HANA

Inbound Scenario

In the inbound scenario, **BTP DMS consumes S/4HANA DMS as an external repository**.

- BTP users can browse, search, and manage documents stored in S/4HANA DMS from within the BTP DMS interface.
- Supported capabilities include:
    - Document creation, editing, and deletion
    - Folder management
    - Metadata and basic/full-text search
    - Rendition viewing
    - Classification and status management

Outbound Scenario

In the outbound scenario, **S/4HANA uses BTP DMS as the content repository**.

- Documents created in S/4HANA (e.g., via CV01N/CV02N or GOS attachments) are stored in **BTP DMS** instead of an on-premise Content Server.
- S/4HANA remains the front-end interface for document creation and retrieval.
- BTP DMS handles storage, lifecycle, and access management for the binary files.

Key point:

- To implement the outbound scenario, **SAP Document Management Service, Integration Option** must be used.
- The Application Option is **not supported** for S/4HANA outbound integration.

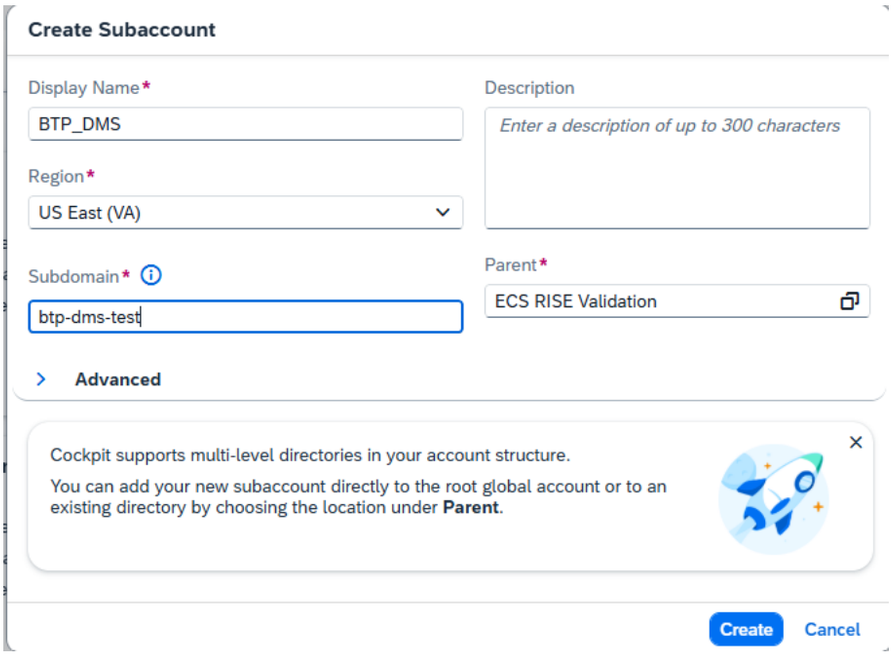
This document focuses on **S/4HANA hosted in ECS (SAP ERP Private Cloud)** and provides detailed guidance for integrating it with **SAP BTP Document Management Service (DMS) Content Server using the Integration Option**. The following sections will outline the step-by-step process for setting up the integration, configuring the repository, and enabling outbound document storage from S/4HANA to BTP DMS.

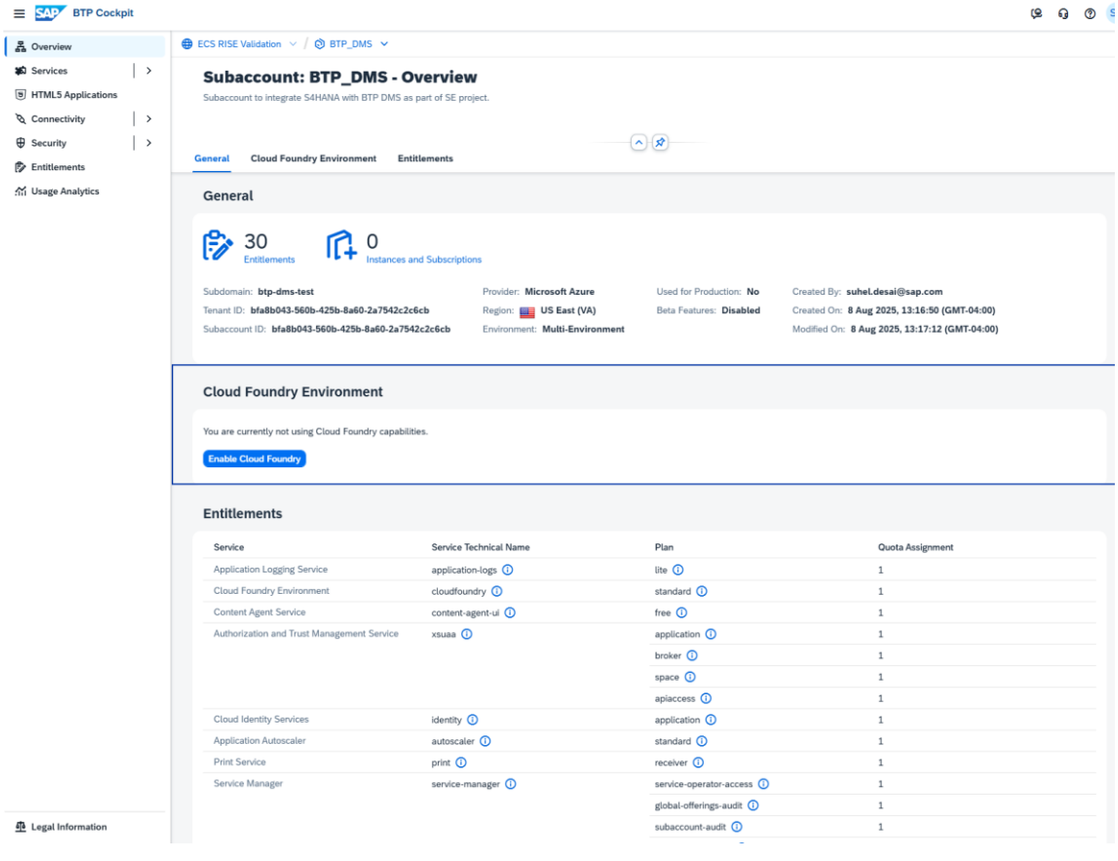
Prerequisites

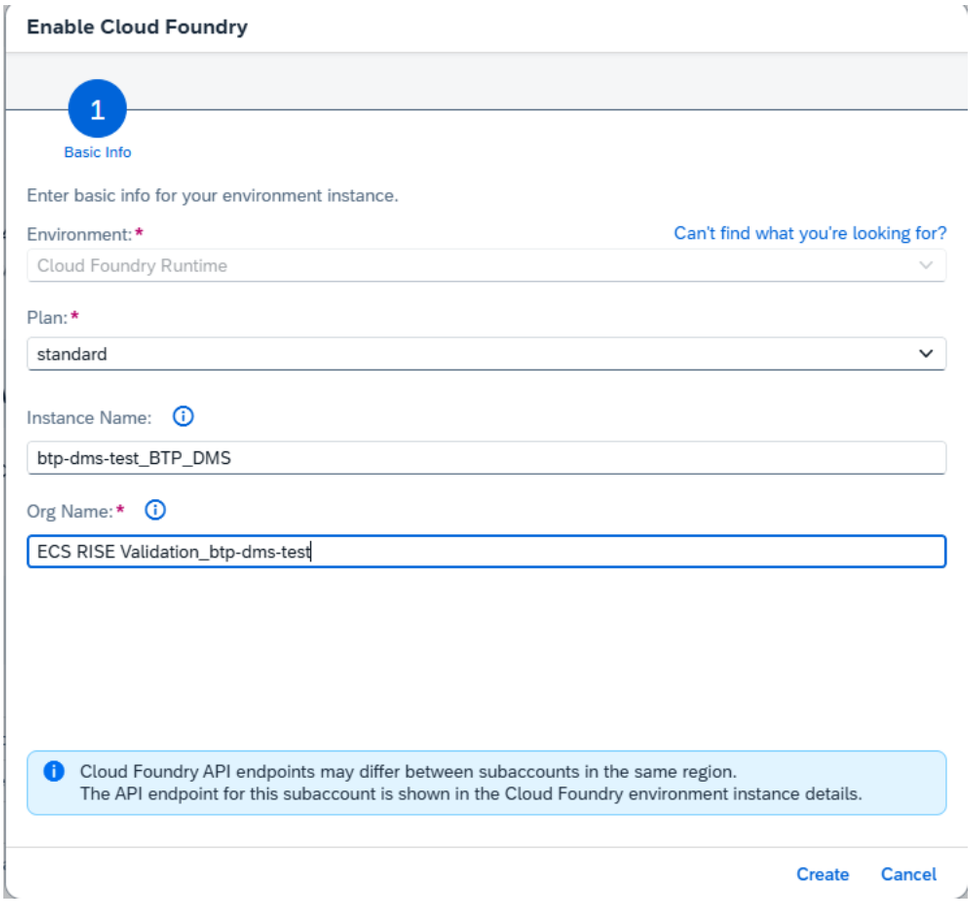
1. You must read and execute the SAP Note - 3595329.
2. You've downloaded the SSL certificates of SAP BTP. For more information about the steps, refer to this knowledge base article KBA-2853519.

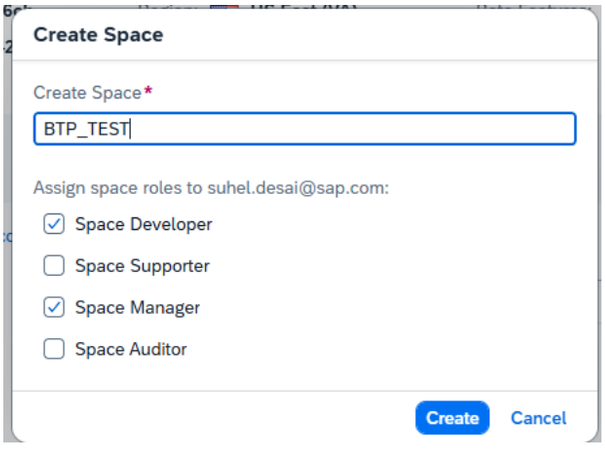
Section 1 – Repository Onboarding

1. Create sub-account



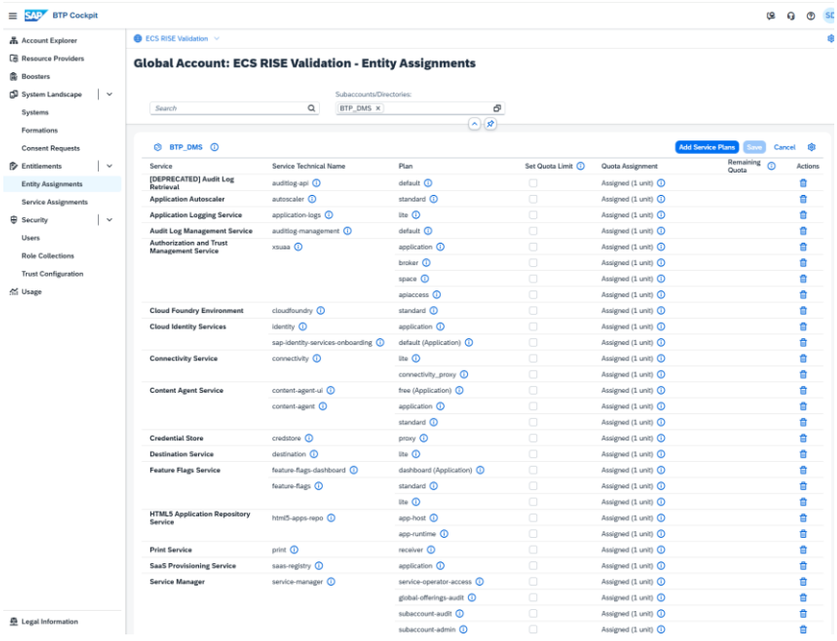




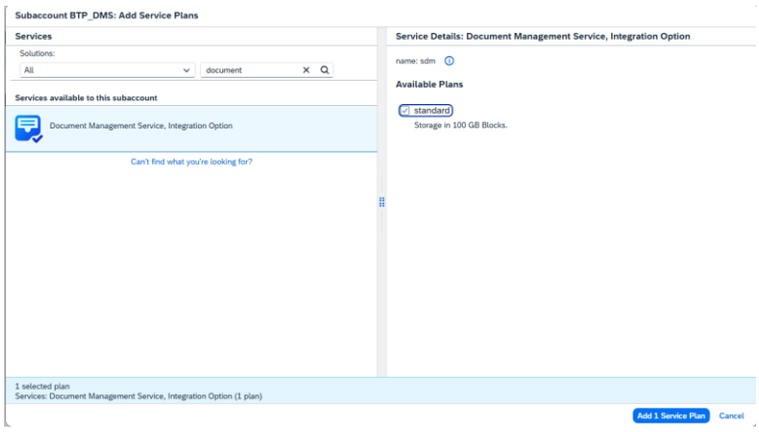


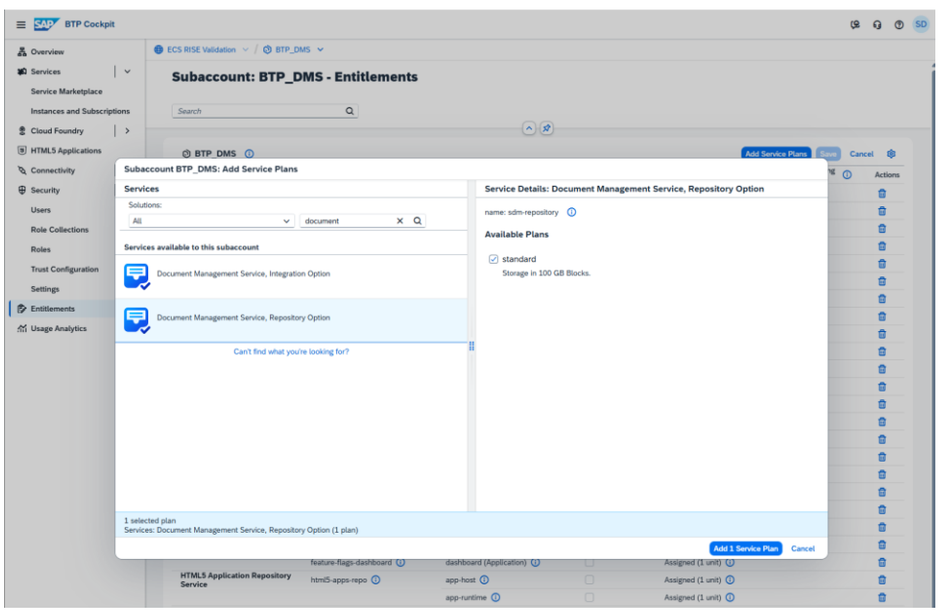
2. Enable Integration Option & Repository Option

1. Add the DMS Integration and repository option service to Sub-account.



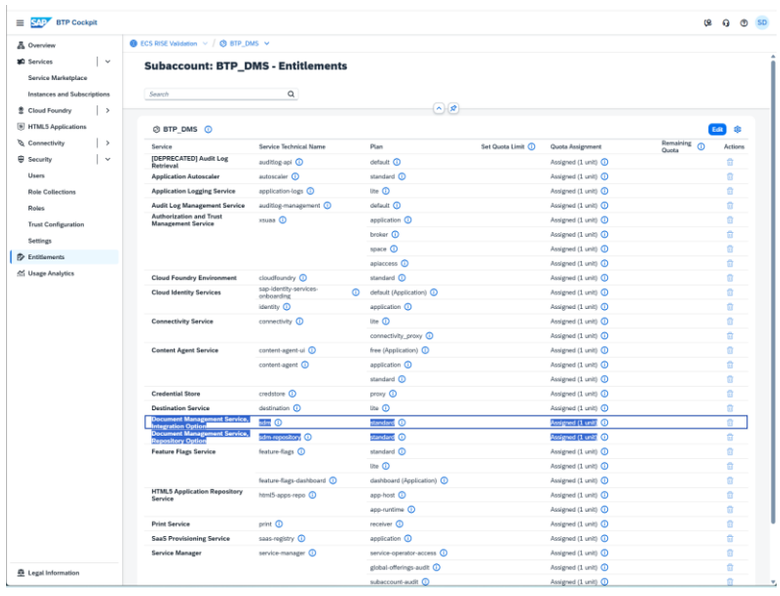
Click Add Service plan and search Document Management Service and select both Integration and repository option





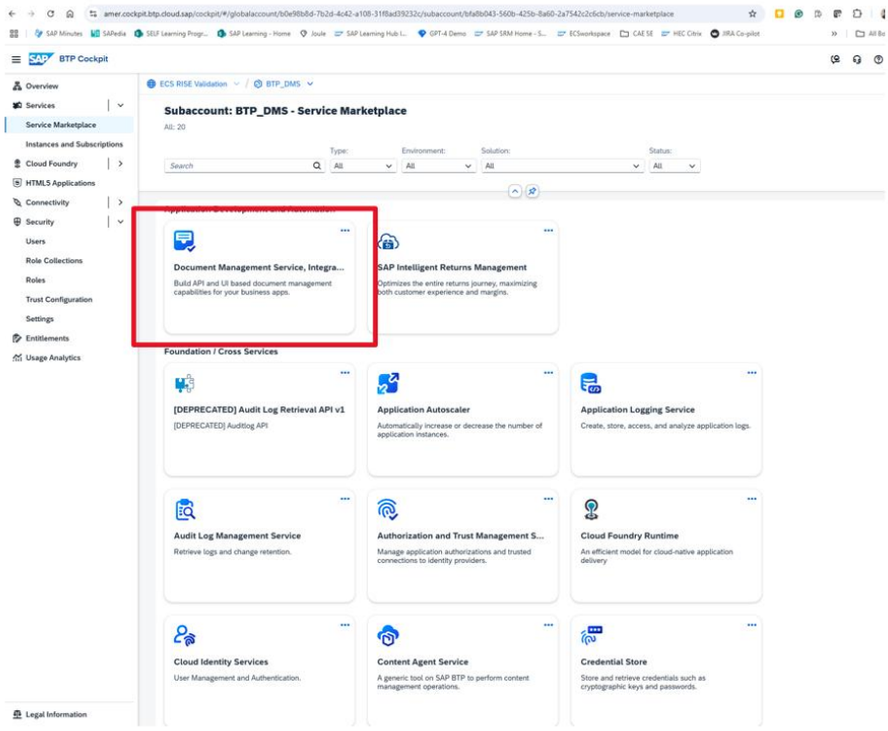
Add and save.

Now verify that both DMS integration and repository option entitlements are active on the subaccount.

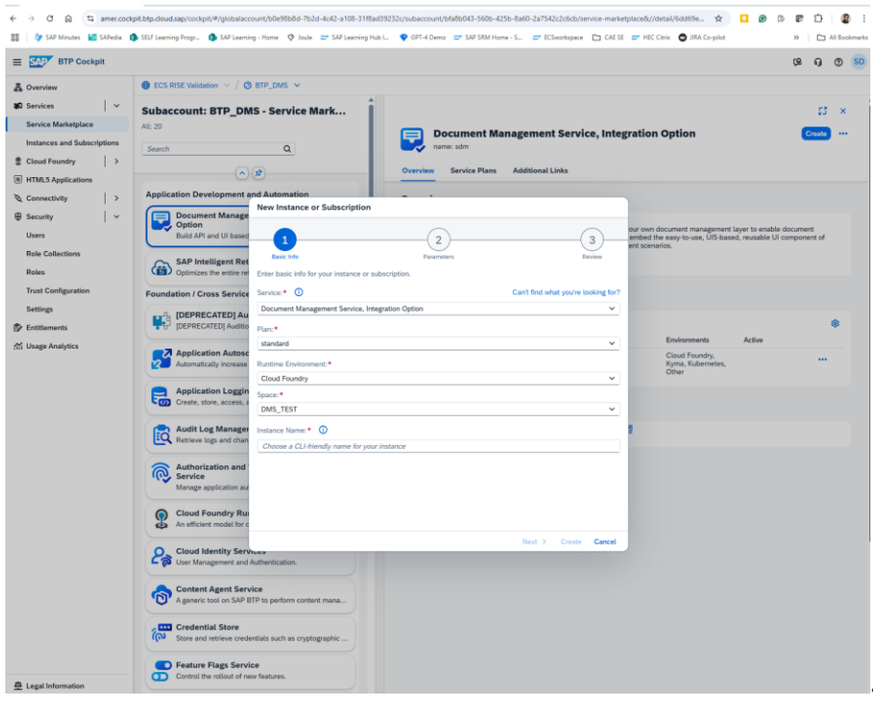


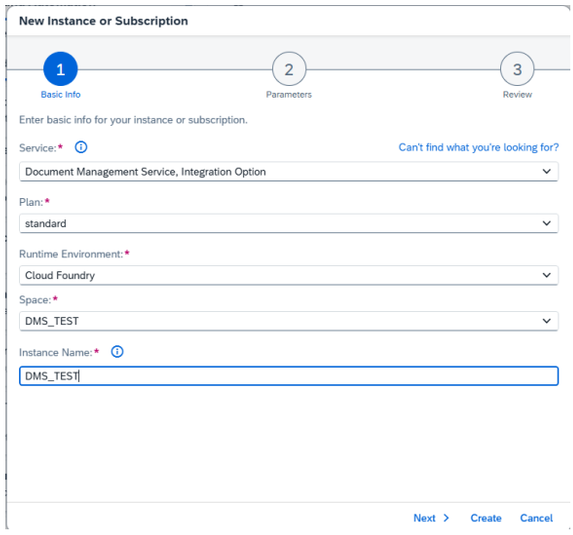
3. Create service instance and service key

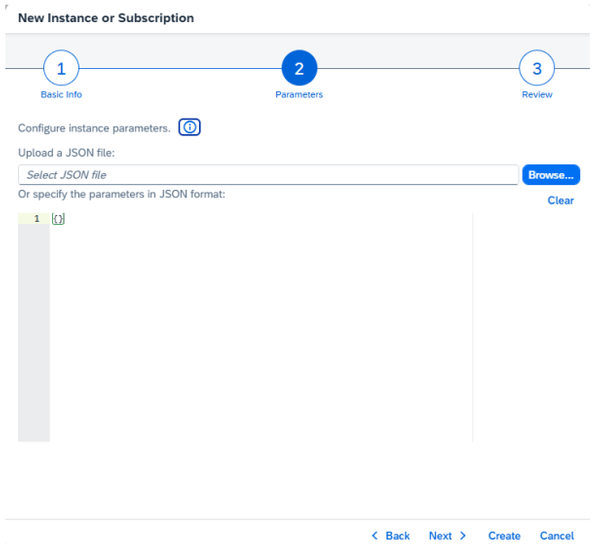
1. Now to create instance, under subaccount go to service marketplace find document management service, integration option

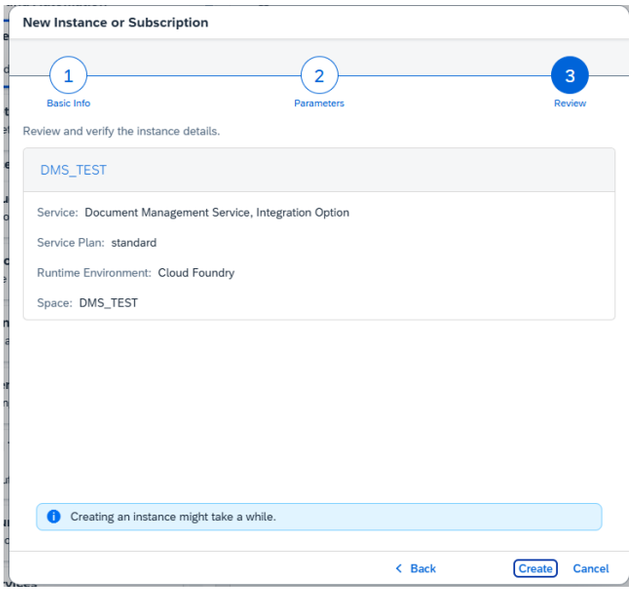


Click Create




New Instance or Subscription
1
Basic Info
2
Parameters
3
Review
Enter basic info for your instance or subscription.
Service:*
Can't find what you're looking for?
Document Management Service, Integration Option
Plan:*
standard
Runtime Environment:*
Cloud Foundry
Space:*
DMS_TEST
Instance Name:*
DMS_TEST
Next
Create
Cancel

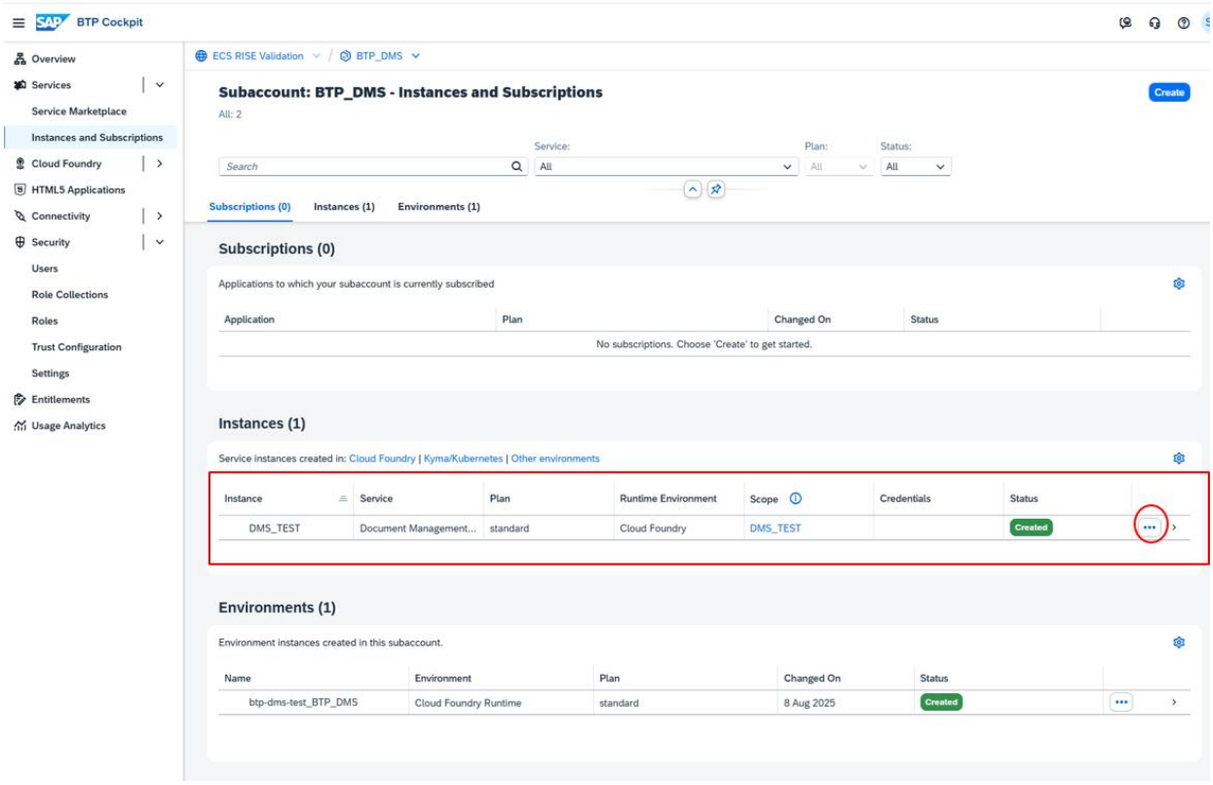

New Instance or Subscription
1
Basic Info
2
Parameters
3
Review
Configure instance parameters.
Upload a JSON file:
Select JSON file
Browse...
Or specify the parameters in JSON format:
Clear
1 {}
Back
Next
Create
Cancel



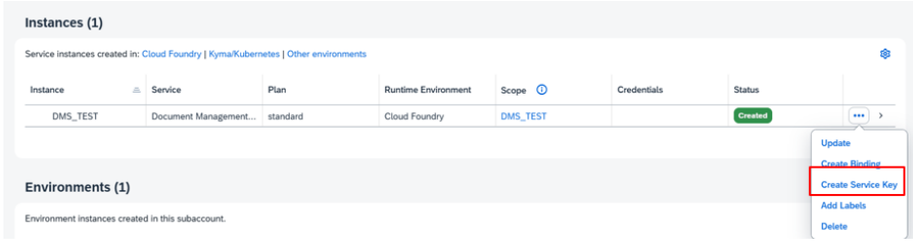
Click Create

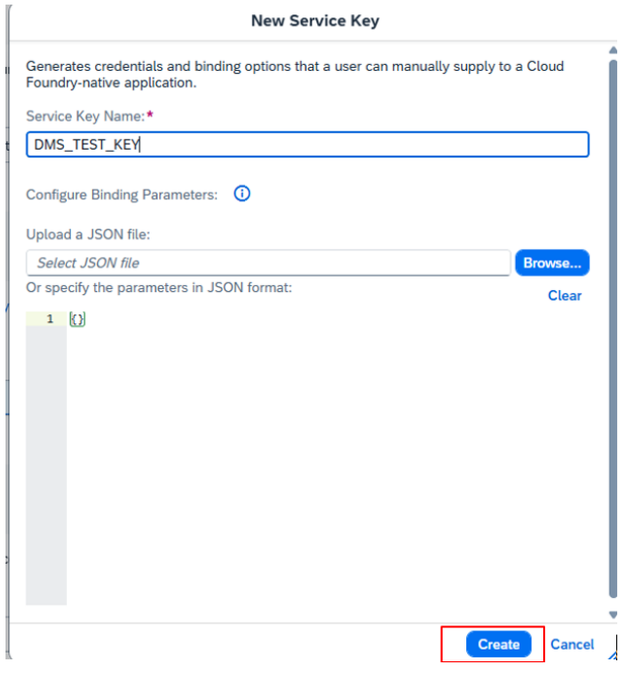
Now, navigate to Instances and subscriptions .

You should see DMS_TEST under instances.

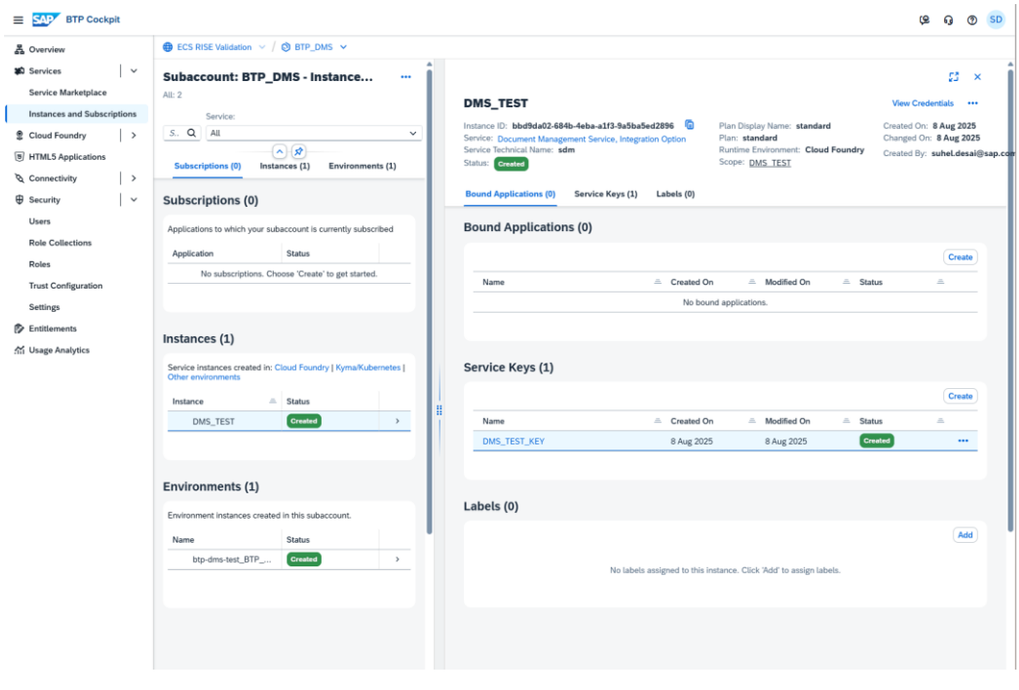


Click 3 dots(...) and create service key

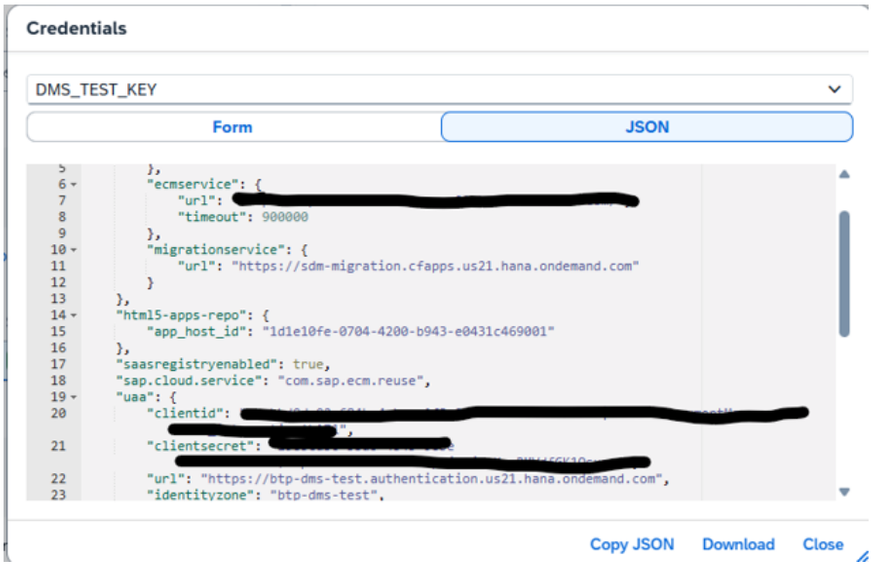




Give Service Key name – In our e.g., it is DMS_TEST_KEY



Click on click on View Credentials to view the key.

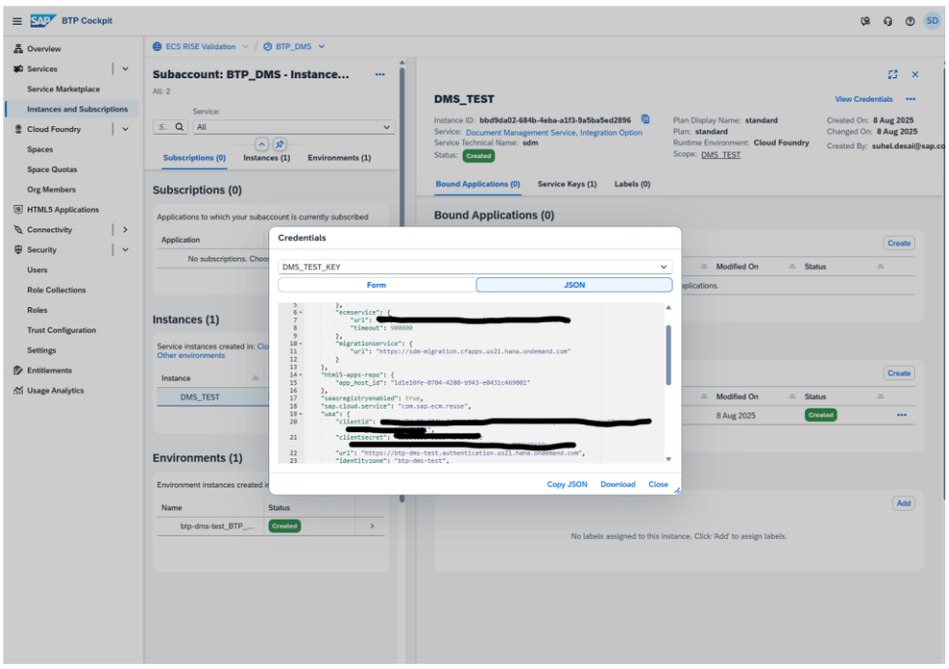


4. Onboarding Repository

Next step is to onboard a repository into Document Management Service, Integration Option instance.

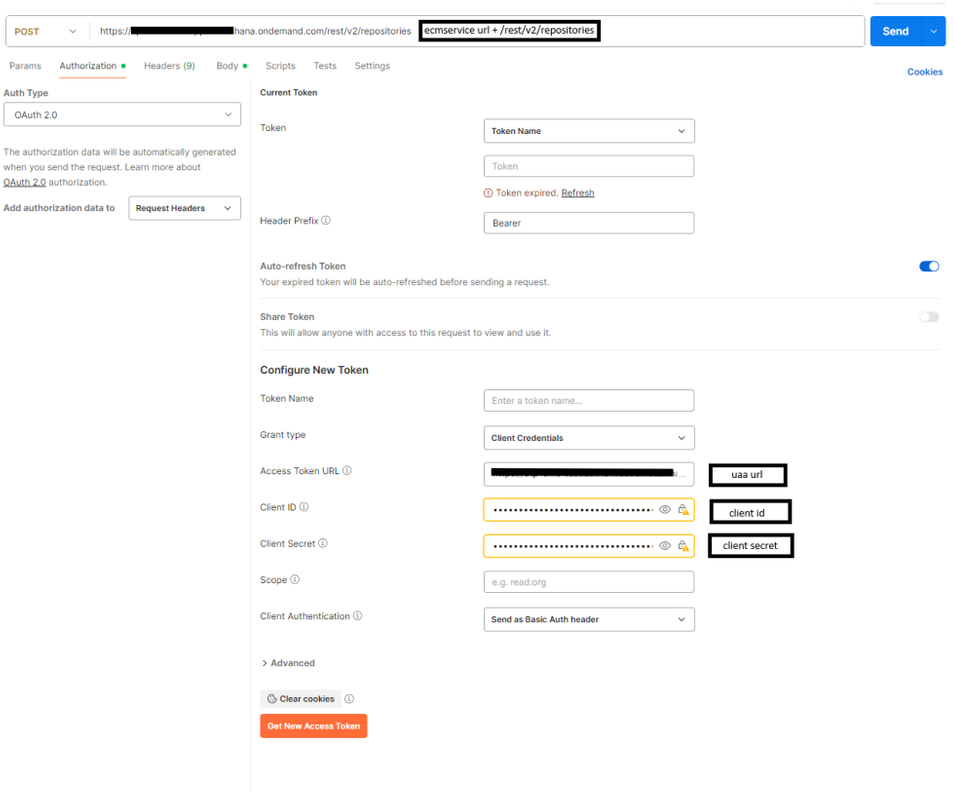
1. Generate a JSON Web token using post man

- Go to DMS_TEST instance created earlier, click on view credentials and note down ecmservice, uaa url, client id, client secret.



In Postman fill the information and generate access token by clicking get new access token button.

Once new token is generated, go to the body section and create JSON in the body section.



Insert following JSON in the body section and click on SEND

```
{
  "repository": {
    "displayName": "ZBTP_DMS_TEST",
    "externalId": "ZBTP_DMS_TEST",
    "isContentBridgeEnabled": "true",
    "description": "Repository for storing S4HANA documents",
    "repositoryType": "internal",
    "cmisRepositoryId": "S4HANA_DOC_REPO",
  }
}
```

*For more such parameters refer help sap page Add your repository using onboarding API. Link in references section

Verify the response as shown below after you hit SEND.

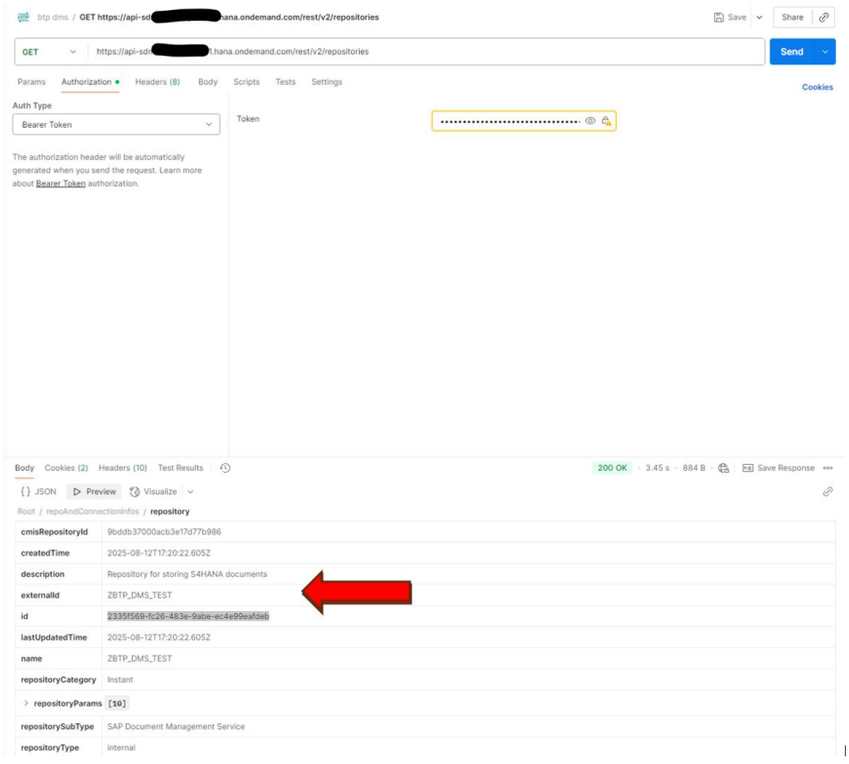
Body Cookies (2) Headers (13) Test Results 201 Created 5.60 s 1.04 KB Save Response

{} JSON Preview Visualize


```
{
    "cmisRepositoryId": "9bddb37800acb3e17d77b986",
    "createdTime": "2025-08-12T17:20:22.605Z",
    "description": "Repository for storing S4HANA documents",
    "externalId": "ZBTP_DMS_TEST",
    "id": "2335f569-fc26-483e-9abe-ec4e99eafdeb",
    "lastUpdatedTime": "2025-08-12T17:20:22.605Z",
    "name": "ZBTP_DMS_TEST",
    "repositoryCategory": "Instant",
    "repositoryParams": [
        {
            "paramName": "changeLogDuration",
            "paramValue": 24
        },
        {
            "paramName": "isVersionEnabled",
            "paramValue": true
        },
        {
            "paramName": "isThumbnailEnabled",
            "paramValue": false
        },
        {
            "paramName": "isVirusScanEnabled",
            "paramValue": false
        },
        {
            "paramName": "hashAlgorithms",
            "paramValue": "SHA-256"
        },
        {
            "paramName": "isContentBridgeEnabled",
            "paramValue": true
        },
        {
            "paramName": "skipVirusScanForLargeFile",
            "paramValue": false
        },
        {
            "paramName": "isEncryptionEnabled",
            "paramValue": false
        },
        {
            "paramName": "isClientCacheEnabled",
            "paramValue": false
        },
        {
            "paramName": "isAIEnabled",
            "paramValue": false
        }
    ],
    "repositorySubType": "SAP Document Management Service",
    "repositoryType": "internal"
```


To confirm repository is created, you may do GET on postman

Note * you need to provide the bearer token generated from earlier step.



5. SAP BTP DMS – Key Endpoints and Usage

These are the core REST and CMIS APIs provided by SAP Document Management Service (DMS), Integration Option. They allow you to onboard and manage repositories, review or delete existing repositories, and browse repository content. Together, these endpoints form the foundation for integrating SAP S/4HANA and other applications with BTP DMS for content storage, navigation, and lifecycle management.

1. Onboard Repository
   POST https://api-sdm-di.cfapps.XX##.hana.ondemand.com/rest/v2/repositories

   When to use: At the start, to register (onboard) a new repository in your subaccount (e.g., connecting S/4HANA DMS or another CMIS-compliant repo).
   Typical use case: Initial setup.

2. Review Repository
   GET https://api-sdm-di.cfapps.XX##.hana.ondemand.com/rest/v2/repositories

   When to use: To list all onboarded repositories and fetch their internal IDs and details.
   Typical use case: Before browsing content or deleting, you first check which repositories exist and retrieve the correct IDs.

3. Delete Repository
   DELETE https://api-sdm-di.cfapps.XX##.hana.ondemand.com/rest/v2/repositories/{repositoryId}

   When to use: To remove a repository that is no longer needed. Uses the internal repository ID (not the external one).
   Typical use case: Cleanup or decommissioning an unused repository.

4. Browse Repository Content
   GET https://api-sdm-di.cfapps.XX##.hana.ondemand.com/browser

   When to use: To navigate the repository's content (folders, documents, metadata).
   This is the CMIS Browser Binding endpoint, used for exploring or integrating with CMIS-compliant tools.
   Typical use case: Reading or browsing existing content once the repository is onboarded.

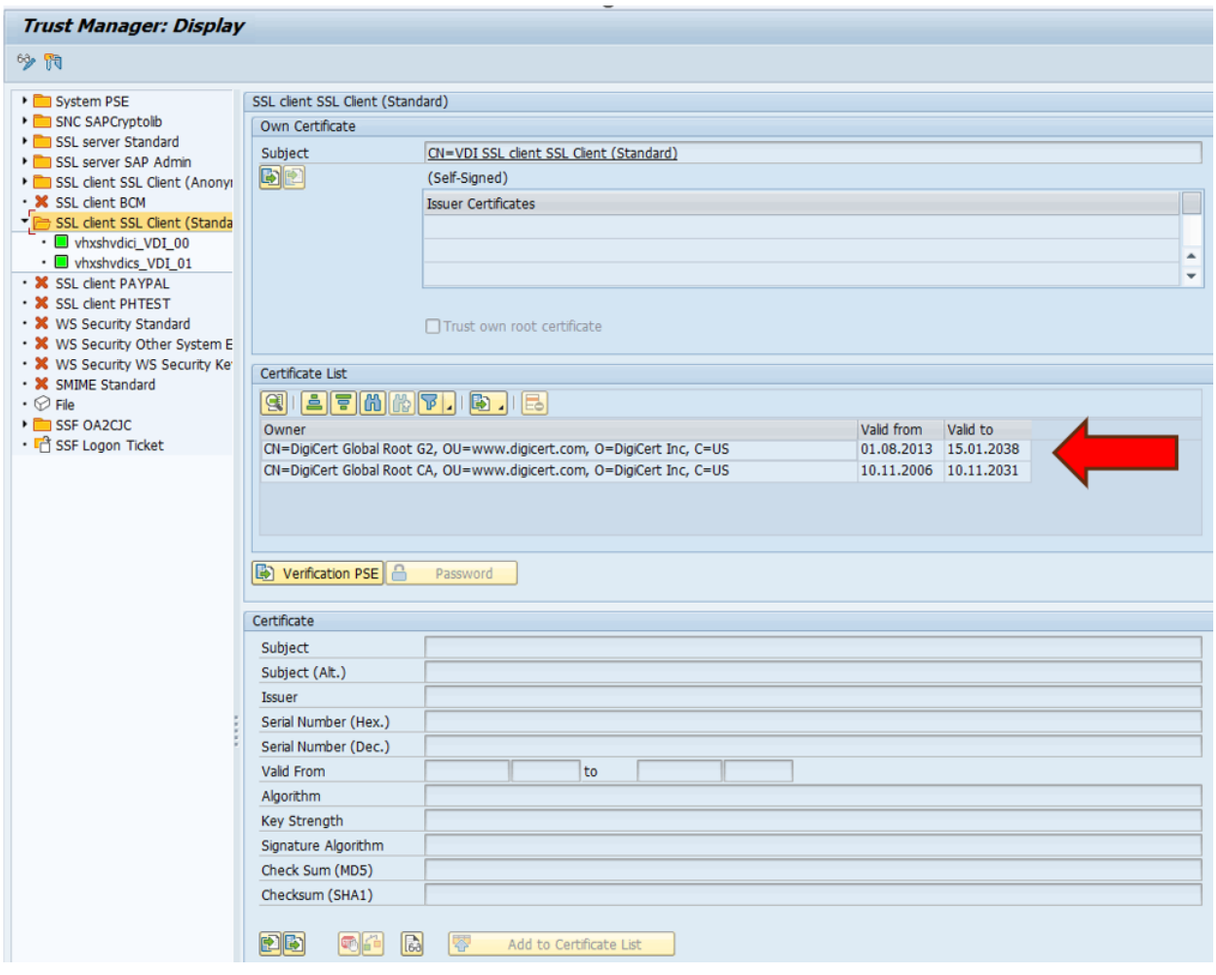
Section 2 : Configuration on S/4HANA

Note* Steps in S/4HANA are performed in the business client.

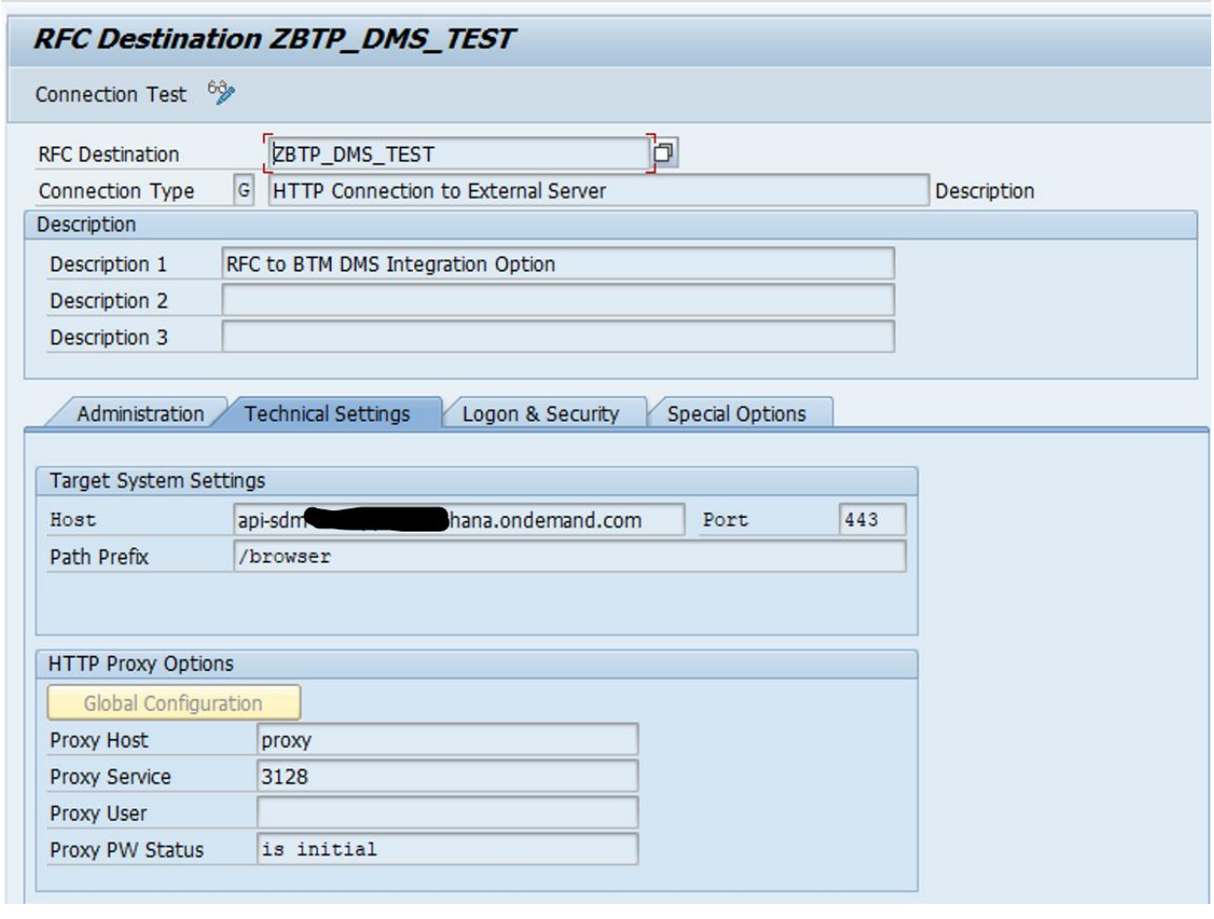
1. Import SSL Certificate in STRUST

Download Root certificates from note 2853519 - How to download the SSL certificate of SAP BTP

Import in Strust Client SSL (Standard)

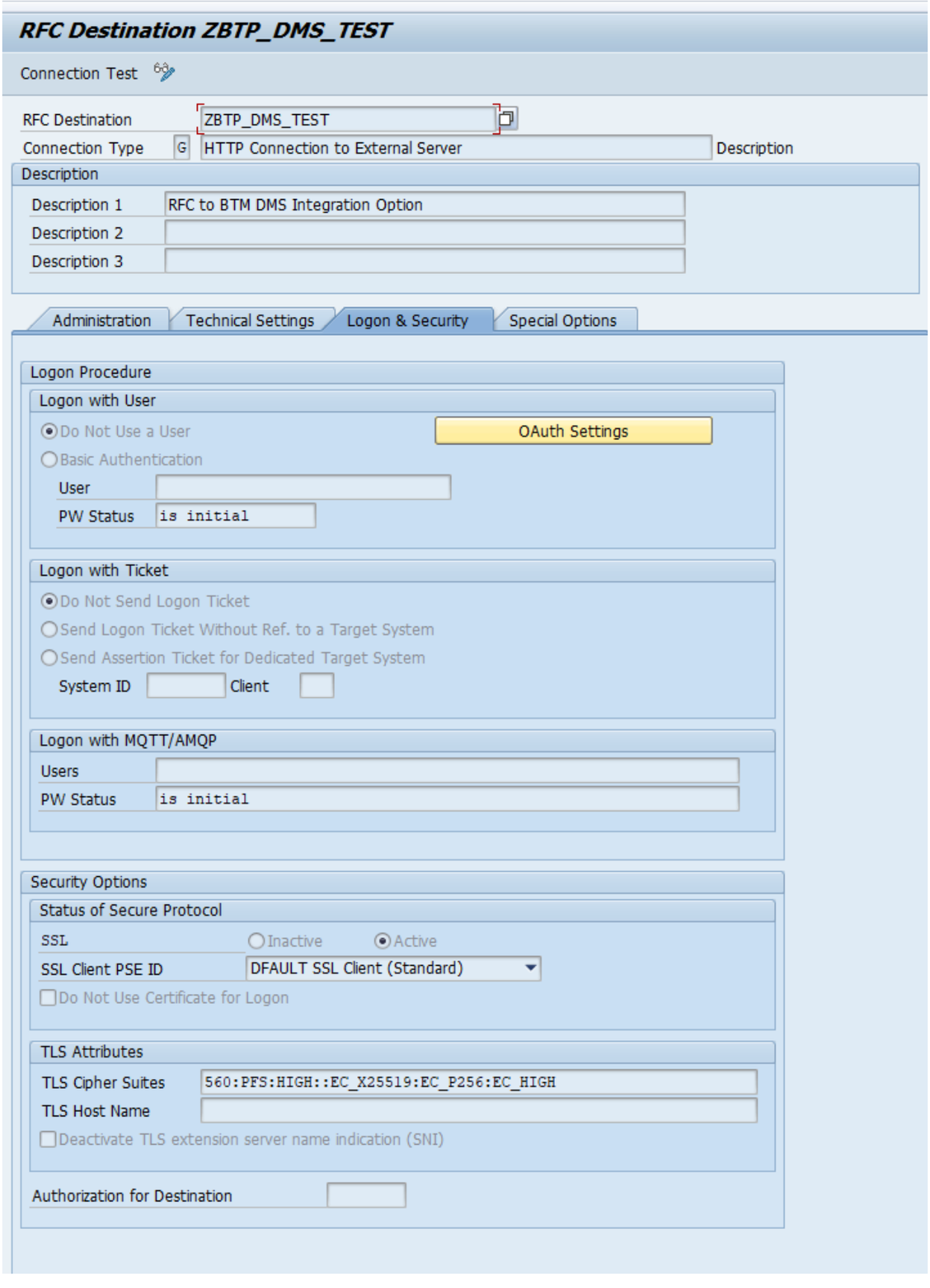


2. Create HTTP RFC



Host = ecs service url
Path Prefix = /browser
Port = 443

Don't forget to add proxy host "proxy" and port "3128" for S/4HANA hosted on SAP ERP private cloud.



Don't put any OAuth settings in the RFC.

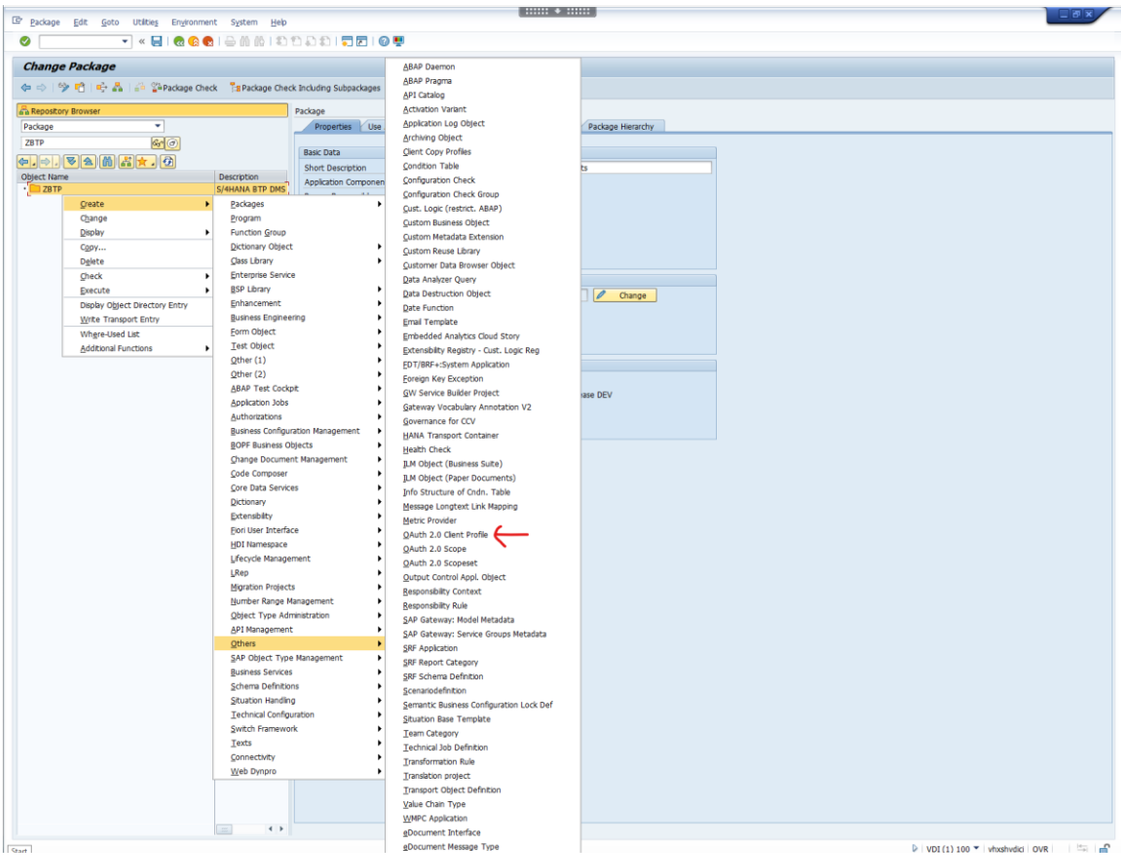
In Security options select DEFAULT SSL Client (Standard) – This is where root certificate was imported earlier. Leave everything else as it is.

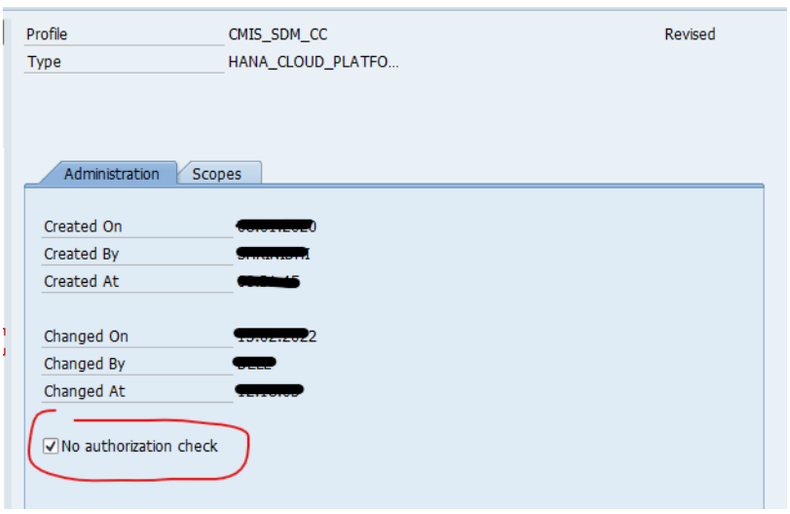
Note* Connection test will ask for user id and password. This is expected behaviour.

3. Create OAuth 2.0 Client Profile

1. Start the object navigator (transaction SE80).
2. Choose Development Object in the drop-down list.
3. To create a development object in the SAP namespace, choose to Create OAuth 2.0 Client Profile in the context menu of the object name.
4. Enter the object name in the Client Profile field(CMIS_SDM_CC)
5. Choose the type of service provider as HANA_CLOUD_PLATFORM.

Save your changes.

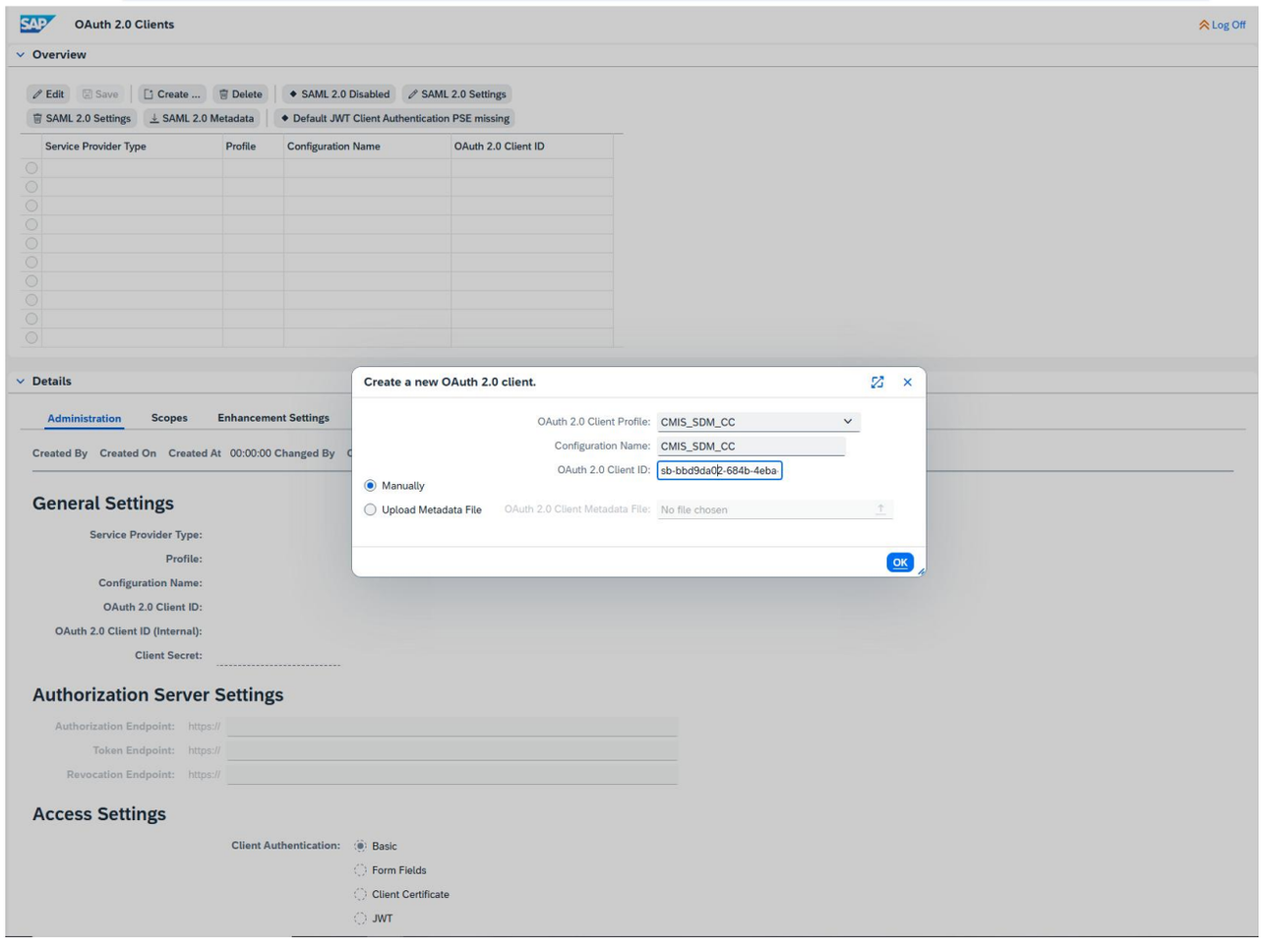




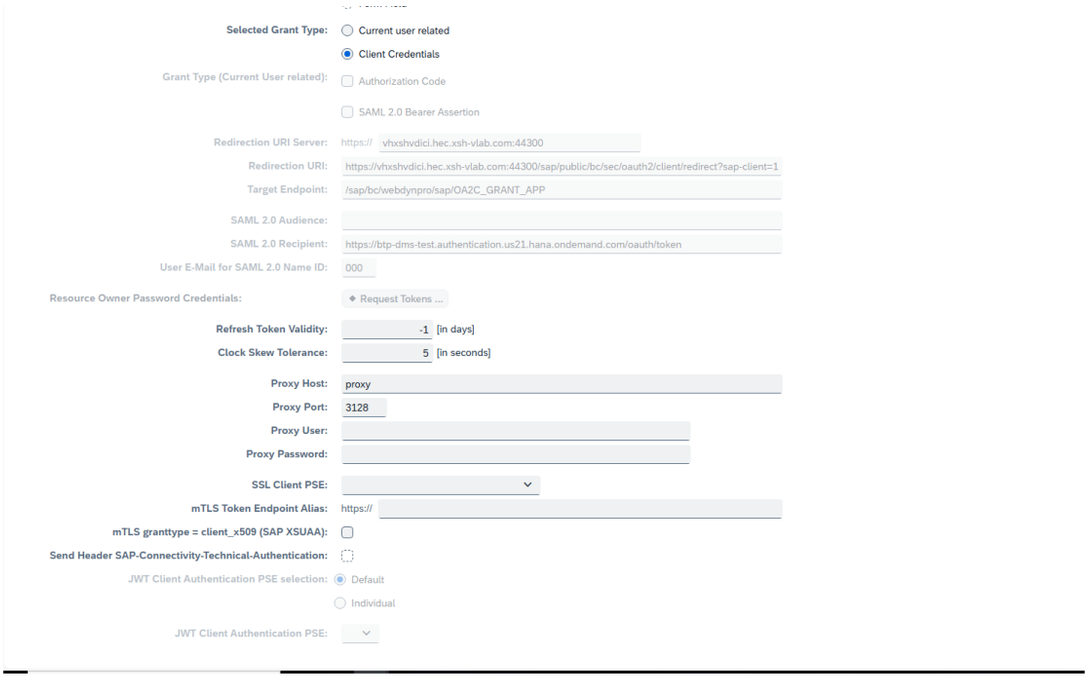
6. Configure OAuth Client

SAP Document Management uses OAuth Authentication. Hence, OAuth client configuration must be done in the SAP S/4HANA backend system.

1. Go to transaction OA2C_CONFIG.
2. Click 'Create'.
3. Choose OAuth Profile created earlier
4. Enter the configuration name of your choice.
5. Enter 'Client Id' from the service key created in Section 1
6. Click 'OK'.

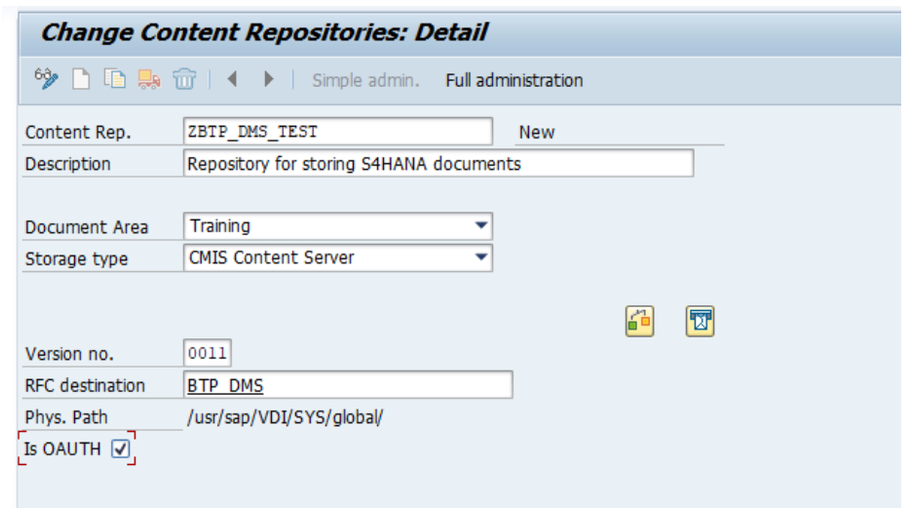


1. Enter 'Client Secret' from the service key created in section 1.3
2. Choose Selected Grant Type as Client Credentials.
3. Enter Authentication Endpoint. It is the "*uaa": "url"* parameter in the service key created in section *1.3* without *'https'*.
4. Enter Token Endpoint. It is the "*uaa": "url"* parameter in the service key with *'/oauth/token' as a* suffix without '*https*'.
5. Enter the Proxy Host: and Proxy Port (ECS private cloud systems on RISE will need to put proxy 3128).
6. Click '*Save*'.

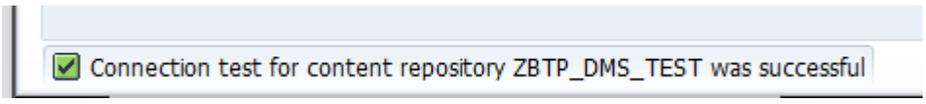


7. Create Content Repository (OAC0)

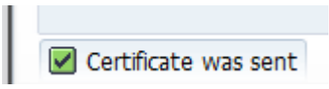
1. Go to transaction OAC0.
2. Click on 'Create'.
3. Give the content repository name the same as the value of the 'externalId' parameter given during the onboarding repository
4. Enter Document Area.
5. Select Storage Type as 'CMIS Content Server'.
6. Enter the RFC destination created in the previous Part.
7. Select the 'Is OAuth' checkbox.
8. Save the configuration
9. Enter the OAuth Profile name.
10. Save the configuration.
11. Do the connection test and check if the connection test is successful in the repository.



Connection test result



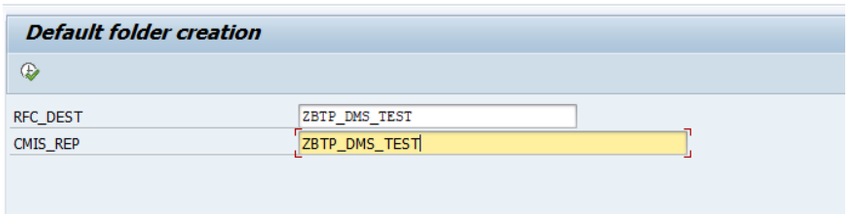
Send RFC result

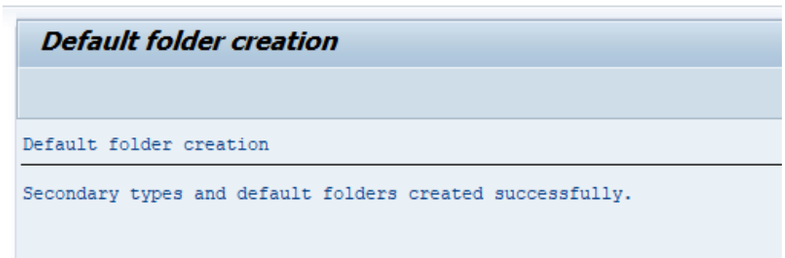


8. Create Secondary Types and Folders

1. Execute Transaction SE38.
2. Enter Report Name 'CMIS_REPOSITORY_SETUP'.
3. Execute report.

Inputs:-
RFC_DEST = RFC destination of the repository created in SM59 CMIS_REP = Content Repository Name created in OAC0

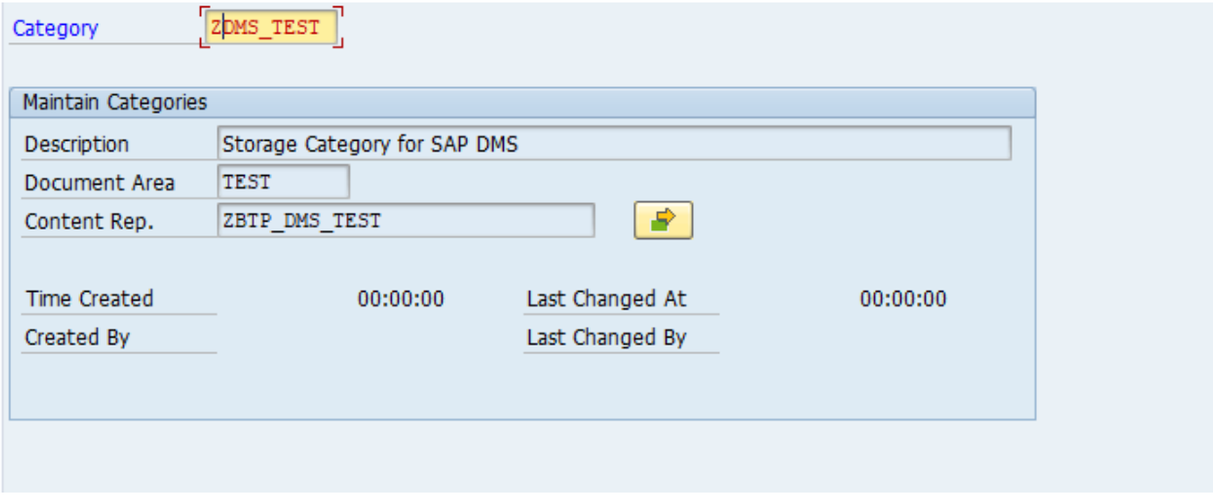




6. Create Storage Category in OACT

1. Go to transaction OACT.

2. Click 'New Entry'.
3. Enter a Category Name.
4. Enter Document Area.
5. Enter the Content Repository Name created in the previous Part.
6. Save Storage Category.



Depending on the document you can create the relevant category and you can now upload documents using CV01/2N or similar tcodes.

3. Migration Approaches

When moving from an on-premise content server to **SAP BTP Document Management Service (DMS) – Integration Option**, or when consolidating multiple repositories, it is essential to migrate not only new documents but also existing content. Without migration, legacy attachments, drawings, or ArchiveLink documents remain tied to the old repository and will be inaccessible once that system is decommissioned.

SAP provides several reports and tools for content migration. The right choice depends on:

- The type of documents (DMS, GOS, ArchiveLink, SRM, CRM, etc.)
- The volume of documents (small, medium, or large-scale)
- Whether metadata updates (links, DIRs, ArchiveLink tables) are required
- The acceptable risk level (delete-after-copy vs. copy-and-verify)

1. RSIRPIRL (SAP Note 389366)

- **Purpose:** Standard SAP report for migrating documents in DMS, GOS, CRM, SRM, and Solution Manager.
- **How it works:**
  - Copies content from the source to the target repository.

    - Can delete originals immediately, or (in newer releases) delay deletion until after confirmation.
- **Advantages:**
    - Delivered in standard SAP, no additional transport required.
    - Works with multiple applications.
- **Limitations:**
    - Logging and error handling are basic.
    - Delay only available in SAP_BASIS > =740. (2991944 - Introducing the Delay mode in report RSIRPIRL)

2. Z_DOC_COPY -SAP Note 2774469, RSCMSCPY2(New) – SAP Note 3080565

- **Purpose:** SAP-delivered custom report option.
- **How it works:**
    - Copies content from source to target repository.
    - Originals are retained until verification is complete.
- **Advantages:**
    - Non-destructive — reduces risk of data loss.
    - Provides detailed logging and restart options.
- **Limitations:**
    - Requires implementation via SAP Note or transport.
    - SAP Application entries need to be adjusted via Custom Code / involving config expert.

3. Z_MIGRATE_ARCHIVELINK -SAP Note 1043676, ARCHIVELINK_MIGRATION(New) – SAP Note 3305771

- **Purpose:** For documents stored using ArchiveLink (e.g., scanned invoices, purchase orders).
- **How it works:**
    - Copies documents to the new repository.
    - Updates ArchiveLink database tables so that business objects point to the new location.
    - Creates new document IDs in the target repository.
- **Advantages:**
    - Specifically designed for ArchiveLink.
    - Ensures links remain intact after migration.
- **Limitations:**
    - Only applicable for ArchiveLink-based content.

4. Other Specialized Reports

- **RSGOS_RELOCATE_ATTA** – Migrates GOS attachments.(Note 1536325)
- **DMS_RELOCATE_CONTENT** – Relocates individual DIRs within DMS. Used in cases for document types e.g. DRW/FOL/SRF etc.)
- **SRM_KPRO_CONTENT_RELOCATION** – Migrates SRM Records/Case Management content.
- **RSCMSEXT / RSCMSEXA / RSCMSIM / RSCMSCPY –** Uses SAPKPROTP (transport files). Suitable for small sets of documents only. Troubleshooting can be difficult, so not recommended for large repositories**.**

Recommendations

- Always perform a pilot migration with a small dataset when possible.
- Use the application-specific tool (ArchiveLink, GOS, or DMS) where available.
- Keep the old repository in read-only mode until the new setup is validated and accepted.
- Perform business-user testing to confirm document access from transactions.
- For performance, parallelize via RSIRPIRL with date ranges (Z_DOC_COPY itself cannot run in parallel unless you trigger with list of documents).
- Maintain detailed migration logs and retain a backup of source repositories until go-live is fully confirmed.
- Incase ICM memory issues, set USEHTTPPLG = OFF in table SDOKPROF so SAPHTTP is used instead of ICM.
- Ensure the latest Content Server patch and SAPHTTP version are applied before migration.

4. Connectivity Troubleshooting Check List

Common Errors

- OAC0 → *Content repository <id> does not exist*
- OAC0 → SSL popup → *SSL client SSL Client (Standard)*
- SM59 → *SSSLERR_PEER_CERT_UNTRUSTED / Create failed*

Fix Steps

☑ Apply all prerequisite SAP Notes listed in note 3595329
☑ Import **DigiCert Global Root G2** certificate into STRUST (SSL Client (Standard) .
☑ Ensure repository onboarding is done with isContentBridgeEnabled=true.
☑ In SM59 Destination:

- Host = https://api-sdm-di.cfapps.<region>.hana.ondemand.com
- No entries in **Logon & Security → OAuth Settings** (delete & recreate if needed).
  ☑ In OAuth client config:
- Token endpoint ends with /oauth/token (not /token2).
- Authorization endpoint = https://<domain>.authentication.<region>.hana.ondemand.com
- Re-paste **Client ID / Secret** from service key.
- Proxy values (if any) must match SM59. For RISE ensure you maintain proxy and 3128.
- Correct *Client Authentication* + *Grant Type* set.

☑ Ignore user/password popup in SM59 test (401 Unauthorized is **expected**).

Success Check

- OAC0 → *Check* → "Connection test was successful."
- OAC0 → *Send RFC info* → "Certificate was sent."

5. References

Blogs

- https://community.sap.com/t5/technology-blog-posts-by-sap/start-with-sap-btp-document-management-service-dms/ba-p/13546912#:~:text=But%20the%20first%20step%20is%20to%20set%20up%20your%20SAP%20BTP%20account%20and%20subscribe%20SAP%20BTP%20DMS%20there
- https://community.sap.com/t5/technology-blog-posts-by-sap/using-document-management-as-a-content-repository-with-sap-s-4hana/ba-p/13485310
- https://community.sap.com/t5/enterprise-resource-planning-blog-posts-by-members/sap-btp-dms-for-s-4-hana/ba-p/13566278
- https://community.sap.com/t5/technology-blog-posts-by-members/demystifying-document-management-service-on-btp/ba-p/13551691
- https://community.sap.com/t5/enterprise-resource-planning-blog-posts-by-sap/content-repository-migration-in-mind/ba-p/13448508
- https://community.sap.com/t5/enterprise-resource-planning-blog-posts-by-sap/content-repository-migration-in-mind-part-ii/ba-p/13449823
- https://community.sap.com/t5/technology-blog-posts-by-sap/content-server-migration-to-sap-document-management-repository/ba-p/13499556

help.sap.com references

- Creating subaccount and space - https://help.sap.com/docs/document-management-service/sap-document-management-service/creating-subaccount-and-space?locale=en-US&version=LATEST
- Creating service instance https://help.sap.com/docs/document-management-service/sap-document-management-service/creating-service-instance-and-service-key?locale=en-US&version=LATEST
- Generate JSON webtoken - https://help.sap.com/docs/document-management-service/sap-document-management-service/generate-json-web-token?locale=en-US&version=LATEST
- Onboarding repository parameters - https://help.sap.com/docs/document-management-service/sap-document-management-service/add-your-repository-using-onboarding-api
- Connecting to BTP DMS Repository Option API https://help.sap.com/docs/document-management-service/sap-document-management-service/connecting-to-document-management-service-repository-option-using-api?locale=en-US&version=LATEST

Notes

https://me.sap.com/notes/3094622
https://me.sap.com/notes/3595329
https://me.sap.com/notes/2774469
https://me.sap.com/notes/389366
https://me.sap.com/notes/1536325